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Editors
Imaging Neuroscience

Dear Editors,

Please consider the Research Article “Geometric Decomposition of Chronic Stroke Registration Asymmetry: Numerical Stability, Registration-Pipeline Sensitivity, and Aphasia Associations in the Aphasia Recovery Cohort” for publication in *Imaging Neuroscience*.

This study addresses a recurring interpretive problem in lesion neuroimaging: registration fields near chronic lesions can be scientifically useful without being physical measurements of tissue displacement or pressure. We introduce a contralateral-normalized, lesional-only deformation proxy, construct a positive-Jacobian stationary log-velocity embedding, and apply an explicitly bounded Helmholtz–Hodge decomposition. Every log-domain case is audited by exponentiating the estimated velocity and comparing the reconstruction with its input deformation.

We additionally repeat the complete analysis using ANTs/SyN forward and inverse mappings with the symmetric MNI152 template. This pipeline-sensitivity analysis includes predefined overlap, Jacobian, and inverse-consistency checks, cross-algorithm descriptor agreement, and identical-split predictive comparisons. Participant-level transforms remain outside the public repository; all aggregate results and a second standalone supplement are regenerated from versioned code.

The study’s central result is a bounded methodological advance rather than an overstated biomarker claim. The log-domain descriptors were numerically valid for every primary case and highly rank-stable across eight perturbations of smoothing, taper, padding, and grid resolution. In 210 participants from the openly available Aphasia Recovery Cohort, deformation and component balance tracked aphasia severity descriptively, but no association survived adjustment for conventional clinical and lesion variables and multiplicity. A small deformation-feature advantage met the within-pipeline prediction criterion in the ANTs/MNI152 analysis but not in SVReg; descriptor agreement and direct same-split comparisons bound this as pipeline-sensitive evidence. Hodge features added no predictive benefit in either pipeline. The complete nonsignificant and sensitivity results are reported in the Supplementary Material. This pattern defines both the reproducible information in the registration geometry and its current limit as an independent behavioral marker.

The work fits *Imaging Neuroscience* because it combines open human neuroimaging, deformable registration, reproducible computational methodology, and out-of-sample brain–behavior modeling. Code, tests, aggregate results, and LaTeX sources are publicly available in the [ARCD deformation analysis repository](#); the source dataset is available from OpenNeuro.

OpenAI Codex assisted with code refactoring, statistical implementation, reproducibility checks, and language editing. The author reviewed the analysis and accepts responsibility for the submitted code and text. The corresponding disclosure appears in the manuscript.

Before submission, the author will confirm originality and exclusive consideration and supply the final funding and competing-interest declarations in the manuscript and submission system.

Sincerely,

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